Atharva Vaidya

AI/ML, Computer Vision Engineer

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Experience

Computer Vision Engineer (contract) May 2024 – Apr. 2025
Carl Zeiss AG, Oberkochen, Germany

- Developed real-time RGB and RGBD-based 3D human pose estimation pipelines for integration into robotic systems
- Built PoC solutions for real-time surgical tool tracking and precise motion capture of human hands
- Generated high-fidelity synthetic data using Mitsuba 3 for accurate instance segmentation in microscopic surgical images
- Implemented generative AI-based solutions and material parameter learning techniques to build a dataset of realistic textures of human iris
- Contributed to the development of a digital twin toolbox based on OpenUSD for opto-robotic simulation and visualization
- Improved a digital shadow of a robotic system with integrated camera for testing and synthetic data generation
- Designed and integrated a conversational language understanding model by Azure AI Language Studio for enabling voice interaction with microscopes

Computer Vision Intern/Master Thesis Dec. 2022 – Dec. 2023
Carl Zeiss AG, Oberkochen, Germany

- Developed solutions for achieving accurate human pose estimation under blanket occlusion using only RGB images
- Extended a rendering pipeline for generating synthetic data with automatic annotations
- Developed tools for manipulating human character models using Blender's Python API
- Implemented domain randomization techniques in the synthetic data to reduce the domain gap and improve generalization to the real data
- Developed training pipelines of CNN and Transformer models like YOLOv7, YOLOv8, ViTPose for synthetic-to-real transfer learning leading to an increase in accuracy by 30 percentage points
- Implemented domain adaptation in YOLOv7 architecture leading to a 3 percentage points improvement in accuracy

Education

Master of Science in Robotic Systems Engineering 2020 - 2024
RWTH Aachen, Germany *Grade: 2.2*
Thesis: Utilizing and Adapting Synthetic Data for Pose Estimation of Occluded Humans in Horizontal Positions

Bachelor of Engineering (Hons.) in Mechanical Engineering 2015 - 2019
BITS Pilani, India *Grade: 8/10*

Skills

Domains: Computer Vision, Synthetic Data, Generative AI, Machine Learning, Natural Language Understanding, Robotics
Programming Languages: Python, C++, C#
Libraries: OpenCV, PyTorch, TensorFlow, Keras, Pytest
Software Tools: Docker, Git, CI/CD, Blender API, ROS
Soft Skills: Problem-solving, Communication, Adaptability, Teamwork
Languages: German (advanced), English (fluent)

Publications

- Peter, R., Oberschulte, E., Wu, E., **Vaidya, A.**, Lindemeier, T., Tagliabue, E., & Mathis-Ullrich, F. (in press). Generation of controllable and photorealistic synthetic cataract surgery images: Blending 3D models and real-world data. In *International Workshop on Simulation and Synthesis in Medical Imaging*. Cham: Springer Nature Switzerland 2025.
- Peter, R., Oberschulte, E., **Vaidya, A.**, Lindemeier, T., Mathis-Ullrich, F., & Tagliabue, E. (2025). Rectification of cornea induced distortions in microscopic images for assisted ophthalmic surgery. Manuscript submitted for publication.